Rajalakshmi Engineering College

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2024_28_III_OOPS Using Java Lab

REC_2028_OOPS using Java_Week 6_CY

Attempt : 1
Total Mark : 40
Marks Obtained : 40

Section 1 : Coding

1. Problem Statement

Bob has been tasked with creating a program using CircleUtils class to calculate and display the circumference and area of the circle.

The program should allow Bob to input the radius of a circle as both an integer and a double and compute both the circumference and area of the circle using separate overloaded methods:

calculateCircumference- To calculate the circumference using the formula 2 * 3.14 * radiuscalculateArea- To calculate the area 3.14 * radius * radius

Write a program to help Bob.

Input Format

The first line of input consists of an integer m, representing the radius of the circle as a whole number.

The second line consists of a double value n, representing the radius of the circle as a decimal number.

Output Format

The first line of output displays two space-separated double values, rounded to two decimal places, representing the circumference of the circle with the integer radius and the double radius, respectively.

The second line displays two space-separated double values, rounded to two decimal places, representing the area of the circle with the integer radius and the double radius, respectively.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 5
3.50

Output: 31.40 21.98
78.50 38.47

Answer

```
import java.util.Scanner;

// You are using Java
class CircleUtils{
   public static double calculateCircumference(int r)
{
      return r*3.14*2;
}
   public static double calculateCircumference(double r)
{
      return r*3.14*2;
}
   public static double calculateArea(int r)
{
      return r*3.14*r;
}
```

```
    public static double calculateArea(double r)

  {
      return r*3.14*r;
  }
}

class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        int radiusInt = scanner.nextInt();
        double radiusDouble = scanner.nextDouble();

        CircleUtils circleUtils = new CircleUtils();

        double circumferenceInt = circleUtils.calculateCircumference(radiusInt);
        double circumferenceDouble =
circleUtils.calculateCircumference(radiusDouble);
        double areaInt = circleUtils.calculateArea(radiusInt);
        double areaDouble = circleUtils.calculateArea(radiusDouble);

        System.out.format("%.2f %.2f\n", circumferenceInt, circumferenceDouble);
        System.out.format("%.2f %.2f", areaInt, areaDouble);

        scanner.close();
    }
}
```

Status : Correct ***Marks : 10/10***

2. Problem Statement

A painter needs to determine the cost to paint different shapes based on their surface area. The program should be designed to handle the area of a sphere and calculate the total painting cost using the following formulas:

Area of sphere: Area = 4 * pi * r² where pi = 3.14Total painting cost: Cost = cost per square meter * area of sphere

The program will consist of three classes:

Shape class: This class should set the shape type and radius.Area class: This class should extend Shape to calculate the area.Cost class: This class should extend Area to calculate the total painting cost.

Input Format

The input consists of a string representing the shape type, a double value representing the radius, and another double value representing the cost per square meter on each line.

Output Format

For a valid shape type of "Sphere":

- The first line prints: "Area of Sphere is: <calculated_area>" rounded to two decimal places.
- The second line prints: "Cost to paint the shape is: <total_painting_cost>" rounded to two decimal places.

For any other shape types, print: "Invalid type".

Refer to the sample output for formatting specifications.

Sample Test Case

Input: Sphere
3.4
5.8

Output: Area of Sphere is: 145.19
Cost to paint the shape is: 842.12

Answer

```
import java.util.Scanner;

// You are using Java
class Shape
{
   protected String shapeType;
   protected double radius;
   public void setShape(String shapeType,Scanner sc)
{
      this.shapeType=shapeType;
```

```
        this.radius=sc.nextDouble();
    }

}
class Area extends Shape
{
    protected double area;
    public void calculateArea()
{
        if(shapeType.equals("Sphere"))
{
            area=4*3.14*radius*radius;
            System.out.printf("Area of Sphere is: %.2f\n",area);
}
        else
{
            System.out.println("Invalid type");
            System.exit(0);
}

}
}
class Cost extends Area
{
    private double costPerSqM;
    public void setCost(double costPerSqM)
{
        this.costPerSqM=costPerSqM;
}
    public void calculateCost()
{
        double totalCost = costPerSqM*area;
        System.out.printf("Cost to paint the shape is: %.2f\n",totalCost);
}

}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        String s = scanner.next();
        Cost shape = new Cost();
        shape.setShape(s, scanner);
```

```
        double costToPaint = scanner.nextDouble();
        shape.calculateArea();
        shape.setCost(costToPaint);
        shape.calculateCost();
    }
}
```

Status : Correct ***Marks : 10/10***

3. Problem Statement

Teena's retail store has implemented a Loyalty Points System to reward customers based on their spending. The program calculates and displays the loyalty points based on whether the customer is a regular or a premium customer.

For regular customers (class Customer), the loyalty points are calculated as:

Loyalty points = amount spent / 10

For premium customers (class PremiumCustomer, which inherits from Customer), the loyalty points are calculated as:

Loyalty points = 2 * (amount spent / 10)

The program should use method overriding for premium customers to calculate their loyalty points. The method that needs to be overridden is calculateLoyaltyPoints in the Customer class.

Input Format

The first line of input consists of an integer representing the amount spent by the customer.

The second line consists of a string representing the premium customer status:

- "yes" if the customer is a premium customer.
- "no" if the customer is not a premium customer.

Output Format

The output should display the loyalty points earned based on the amount spent and the customer type.

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 50
yes

Output: 10

Answer

```
import java.util.Scanner;
class Customer{
   int calculateLoyaltyPoints(int r)
{
      return r/10;
}
}
class PremiumCustomer extends Customer
{
   int calculateLoyaltyPoints(int r)
{
      return 2*(r/10);
}
}
public class Main {
   public static void main(String[] args) {
      Scanner scanner = new Scanner(System.in);

      int amountSpent = scanner.nextInt();

      String isPremium = scanner.next().toLowerCase();

      Customer customer;

      if (isPremium.equals("yes")) {
         customer = new PremiumCustomer();
      } else {
         customer = new Customer();
```

```
    }
    int loyaltyPoints = customer.calculateLoyaltyPoints(amountSpent);

    System.out.println(loyaltyPoints);
  }
}
```

Status : Correct ***Marks : 10/10***

4. Problem Statement

Teena is launching a new airline, Boeing747, and needs to calculate the total revenue generated from ticket sales based on the ticket cost and seat availability. Teena's airline offers two types of seats: regular and premium. The ticket cost and seat availability for both types of seats need to be considered for revenue calculation.

To help with this, Teena wants to implement a system using multilevel inheritance with three classes:

Airline: This class will have the ticket cost as an attribute and defines the method setCost(double cost) and double getCost().Indigo: This class will extend Airline and add the seat availability attribute and defines the method getSeatAvailability() and setSeatAvailability(int seatAvailability) .Boeing747: This class will extend Indigo and include a method calculateTotalRevenue() based on the ticket cost and seat availability .

Teena needs to calculate the total revenue using the formula:

Total Revenue = ticket cost * seat availability

Help Teena implement this system for calculating the revenue of her airline.

Input Format

The first line of input consists of a double value, representing the flight's ticket cost.

The second line consists of an integer, representing seat availability.

Output Format

The first line of output prints "Ticket Cost: Rs. " followed by a double value representing the ticket cost rounded to one decimal place.

The second line of output prints "Seat Availability: X seats" where X is an integer value representing the seat availability.

The third line of output prints "Total Revenue: Rs. " followed by a double value representing the total revenue rounded to one decimal place.

Refer to the sample output for the exact text and format.

Sample Test Case

Input: 1000.0
100

Output: Ticket Cost: Rs. 1000.0
Seat Availability: 100 seats
Total Revenue: Rs. 100000.0

Answer

```
import java.util.Scanner;

class Airline
{
  private double cost;

  public void setCost(double cost)
{
    this.cost = cost;
}
  public double getCost()
{
    return cost;
}
}
class Indigo extends Airline
{
  private int seatAvailability;
  public void setSeatAvailability(int seatAvailability)
```

```
{
    this.seatAvailability = seatAvailability;
}
  public int getSeatAvailability()
{
    return seatAvailability;
}
}
class Boeing747 extends Indigo
{
  public double calculateTotalRevenue()
{
    return getCost() * getSeatAvailability();
}


}
public class Main {
  public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    Boeing747 plane = new Boeing747();

    double ticketCost = scanner.nextDouble();
    plane.setCost(ticketCost);
    int seatAvailability = scanner.nextInt();
    plane.setSeatAvailability(seatAvailability);

    System.out.printf("Ticket Cost: Rs. %.1f\n", plane.getCost());
    System.out.println("Seat Availability: " + plane.getSeatAvailability() + "
seats");
    System.out.printf("Total Revenue: Rs. %.1f\n",
plane.calculateTotalRevenue());
  }
}
```

Status : Correct ***Marks : 10/10***